



Published in final edited form as:

Int J Cancer. 2018 December 01; 143(11): 2767–2776. doi:10.1002/ijc.31821.

Circulating Inflammatory Markers and Colorectal Cancer Risk: A Prospective Case-cohort Study in Japan

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Abstract

Blood levels of inflammation-related markers may reveal molecular pathways contributing to carcinogenesis. To date, prospective associations with colorectal cancer (CRC) risk have been based on few studies with limited sets of analytes.

We conducted a case-cohort study within the Japan Public Health Center-based Prospective Study Cohort II, comparing 457 incident CRC cases during median 18 years follow-up with a random subcohort of 774 individuals. Baseline plasma levels of 62 cytokines, soluble receptors, acute-phase proteins, and growth factor markers were measured using Luminex bead-based assays. We estimated hazard ratios (HRs) associating each marker with CRC risk by Cox proportional hazards models adjusted for potential confounders. Sub-analyses compared cases by years after blood draw (< 5 vs. ≥ 5) and anatomical subsite (colon vs. rectum).

Linear trends in quantiles of four C-C motif ligand (CCL) chemokines, one C-X-C motif ligand (CXCL) chemokine, and a soluble receptor were nominally associated with CRC risk based on $P_{trend} < 0.05$, but none met false discovery rate corrected statistical significance. HRs for the 4th vs. 1st quartile were: 1.69 for CCL2/MCP1, 1.61 for soluble tumor necrosis factor receptor 2, 1.39 for CCL15/MIP1D, 1.35 for CCL27/CTACK, 0.70 for CXCL6/GCP2 and 0.61 for CCL3/MIP1A. Among cases diagnosed 5+ years after enrollment, CCL2/MCP1, CCL3/MIP1A and CXCL6/GCP2 retained nominal statistical significance. There were no significant differences in associations between colon and rectum.

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DISCLOSURE OF POTENTIAL CONFLICTS OF INTEREST

No potential conflicts of interest were disclosed.

Our findings implicate chemokine alterations in colorectal carcinogenesis, but require replication for confirmation. Noninvasive chemokine assays may have potential application in colorectal cancer screening and etiologic research.

Keywords

Colorectal Cancer; Biomarker; Inflammation; Chemokines

INTRODUCTION

Colorectal cancer (CRC) is the third most common cancer in the world, which accounted for almost 1.4 million new cases and 694,000 deaths in 2012 ¹. No single predominant risk factor attributes to CRC, but family history of CRC, smoking, excessive alcohol consumption, high intake of red and processed meat, obesity, physical inactivity and diabetes have been identified as risk factors, apart from age and male sex, in epidemiologic studies ^{2, 3}.

Inflammatory bowel disease, such as ulcerative colitis and Crohn's disease, is an established risk factor for CRC, suggesting link of chronic inflammation with malignant disease ⁴. This is also supported by lowered risk associated with regular use of aspirin and other non-steroidal anti-inflammatory drugs (NSAIDs), seen in observational studies and randomized clinical trials ^{5, 6}, and in studies reporting that polymorphisms in inflammation genes are associated with CRC risk ^{7, 8}.

Few prospective studies have investigated the association of circulating inflammatory markers with CRC risk. Studies addressing circulating inflammatory markers were confined to a few candidate markers ⁹. However, even the most frequently studied marker, C-reactive protein (CRP), shows inconsistent results across studies. Similarly, evidence for an association between other inflammatory biomarkers, such as interleukin-6 (IL-6), TNF-alpha (TNF- α) and risk of CRC is inconclusive ⁹. However, these independently studied markers represent only a subset of the inflammation process, which is a complex interplay of host and the environment affecting various stages of tumor development, progression as well as response to therapies ^{4, 10, 11}.

New technologies have enabled some studies to utilize multiplex panels, which gives a broader view of the soluble marker signatures, to explore the role of inflammation in some diseases ¹². However, no prior studies have comprehensively analyzed inflammatory markers with CRC risk.

Discovery of inflammatory biomarkers associated with CRC will provide insights to the understandings of carcinogenic process, and also improve our knowledge for potential usages in prevention and therapeutic intervention. Thus, in the current study, we investigated the association of CRC with circulating inflammatory markers, including various chemokines, cytokines and growth factors on a qualified multiplex panel in a prospective cohort study in Japan.

METHODS

Study Population

The Japan Public Health Center-based Prospective Study (JPHC Study) Cohort II is an ongoing cohort study initiated in 1993-1994, investigating cancer, cardiovascular and other lifestyle-related diseases in the general population¹³. It enrolled 78,825 Japanese citizens, aged 40-69 years at baseline, in 6 prefectural public health center areas (Ibaraki, Niigata, Kochi, Nagasaki, Okinawa, and Osaka) across the country. Self-administered questionnaire data on personal medical history and lifestyle factors (including smoking, alcohol and physical activity), family history of colorectal cancer and anthropometric measurements were collected at baseline.

A total of 23,335 subjects responded to the questionnaire and donated 10 mL of blood samples during health checkups in 1993-1995. A subcohort of 774 subjects was selected by stratified random sampling (with sampling fractions for ages <50, 50-59, 60-69 and 70-79 years, respectively, of 2%, 4%, 8% and 8% for males, and 1%, 2%, 4% and 4% for females) to approximate the age and sex distribution of cancer cases.¹⁴ The detailed exclusion criteria and numbers are shown in Figure 1. We followed subjects through 31 December 2010 and identified 457 newly diagnosed CRC cases (23 in subcohort, 434 in non subcohort). CRC cases were identified through voluntary reports from major local hospitals in the study areas, from the data linkage with population-based cancer registries, supplemented by death certificate data provided by the Ministry of Health, Labor and Welfare. CRC cases were defined using International Classification of Diseases for Oncology, 3rd edition (ICD-O-3) by WHO (C18.0-C20.9)¹⁵.

The study protocol was ethically reviewed and approved by the National Cancer Center, Japan (IRB No. 2004-059).

Laboratory Methods

Blood specimens were collected at the time of their health check-ups into vacutainer tubes containing heparin. The samples were centrifuged within 12 hours and divided into plasma and buffy layers, which were biobanked at -80°C. Vials for this analysis had previously undergone two freeze-thaw cycles for prior serologic measurements.

Circulating levels of 67 inflammatory and immunity markers were measured in plasma samples, using a previously developed and validated set of multiplex panels of Luminex bead-based assays (EMD-Millipore Inc, Billerica, MA)¹⁶. Samples were tested neat for the three Cytokine Panels, at 1:5 dilution for the Soluble Receptor Panel, at 1:2000 dilution for the Cardiovascular Disease Panel, and at 1:2 dilution for the High Sensitivity T Cell Panel (representing a final 1:4 dilution due to the internal assay dilution of 1:2). All the panels were tested according to the manufacturer's recommend protocol for same day testing, except for the High Sensitivity T Cell Panel which was incubated overnight. Concentrations were estimated by interpolation on either a four- or five-parameter logistic curve fitted to measurements of a standard serially diluted at least seven times. Case and subcohort samples were randomly interspersed on test plates. To monitor reproducibility within and between plates, a set of masked samples (total N=56) were tested in three wells, in two random wells

on one plate and a third well on a different plate. Two markers (IL3, TNFB) detectable in less than 10% of samples were excluded from analysis. Three markers (IL7, CXCL8/IL8 and CCL20/MIP3A) were measured with both conventional and high-sensitivity assays in different multiplex panels. Since IL7 was more frequently detectable with the high-sensitivity (73%) than conventional (37%) assay, the high-sensitivity results were used in statistical analysis. On the other hand, CXCL8/IL8 and CCL20/MIP3A were more frequently detectable with the conventional assays (both >90%), so these were the results retained for analysis. Log-transformed marker levels in the replicate samples had median 13% overall coefficient of variation (CV) (all but three <30%) and median 90% intra-class correlation (ICC) (all but six > 70%) (Supporting Information TableS1).

Statistical Analyses

Given the large dynamic range (up to 4 orders of magnitude), markers were analyzed as quantiles for a more robust analysis. Levels were categorized based on the distribution among members of the subcohort, with the lowest quantile used as the reference group. To accommodate measurements less than the lower limit of detection (LLOD), we categorized markers *a priori* as follows: markers with < 25% of individuals below LLOD were categorized into quartiles (N=48 markers); markers with 25% to 50% of individuals below the LLOD were categorized as less than LLOD and tertiles of detectable markers (N=10 markers); markers with 50% to 75% of individuals below the LLOD were categorized into three groups, less than LLOD and below and above the median of the detectable marker values (N=0 markers); markers with 75-90% individuals below the LLOD were categorized as undetectable and detectable (N=4 markers).

Cox proportional hazards regression, stratified by age-group and sex was used to calculate the hazard ratios (HRs) and 95% confidence intervals (CIs) for the association of each marker with CRC risk with robust variance adjustment to account for the case-cohort design^{17, 18}. Age was used as the model time metric. For individuals in the sub-cohort follow-up started at age at blood draw (entry) and ended at age at CRC diagnosis or censoring, defined as age at death, emigration from the study area or at last date of follow-up date, December 31st, 2010, which ever came first. For CRC cases that were non-subcohort, follow-up started 60-days prior to CRC diagnosis and ended at age at CRC diagnosis. Thus these cases contributed information only to their own risk set¹⁷. Sensitivity analyses using shorter time windows of 1 or 30 days did not alter the results.

A fully adjusted model includes family history of CRC (yes or no), history of diabetes (yes or no), BMI category (< 21, 21-22.9, 23-26.9, ≥27 kg/m²), smoking (never, past, current [< 20 and ≥ 20 cigarettes/day]), alcohol drinking (no or occasionally drinking, currently drinking almost every day [1-150, 151-300 and ≥ 300 g ethanol/week]), physical activity (quartiles of metabolic equivalents/day). Furthermore, the quantiles were analyzed as ordinal variables to investigate the linear trend of per quantile increase of the marker levels and CRC risk. We present results in detail only for those markers with $P_{trends} < 0.05$. False discovery rate (FDR) p-values were also calculated to adjust for multiple testing.

Spearman partial correlations were calculated for the significant markers ($P_{trends} < 0.05$), with age, sex and other confounders used in the Cox model. Furthermore, all selected

markers were also included jointly in an adjusted Cox model. We also performed stratified analyses by latency (time from blood collection to diagnosis of < 5 years *vs.* ≥ 5 years) and anatomic subsite (colon *vs.* rectum). $P_{interaction}$ for the latency analysis was calculated by testing for equality between the two time intervals of risk associations per change in marker quantile, censoring at 5 years for the earlier interval and having follow-up start at 5 years for the later interval. $P_{heterogeneity}$ for the difference between colon *vs.* rectum was calculated by the duplication method as the Wald type chi-square p-value for the interaction between subsite and marker quantile ¹⁹.

All tests of statistical significance were based on two-sided p-values < 0.05 and were carried out in SAS version 9.3 (SAS Inc, Cary, NC).

RESULTS

Baseline characteristics of subcohort members and cases are presented in Table 1. The median time from blood collection to CRC diagnosis was 9.5 years (interquartile range = 4.6 – 13.2 years). Compared with subcohort members, the CRC cases had higher percentage of family history of CRC (2.4%) and history of diabetes (7.9%). Among CRC cases 56% were diagnosed at early stage.

Figure 2 shows all the markers and their associations with CRC risk per quantile increase, adjusted for all potential confounders. A list of the six markers with nominal $P_{trend} < 0.05$ are shown in Table 2. These six markers include four C-C motif ligand (CCL) chemokines, one C-X-C motif ligand (CXCL) chemokine, and a soluble tumor receptor. Higher quantiles of CCL2/MCP1, CCL15/MIP1D, CCL27/CTACK, and sTNFR2 were significantly associated with higher risk of CRC, whereas increases of CCL3/MIP1A, CXCL6/GCP2 were associated with lower risk. However, the P_{trends} did not remain significant when FDR corrections were applied.

The associations of each quantile category of the six associated markers are presented in Table 2. For four of the six markers, the elevated levels were associated with statistically significant increased CRC risk. The association for the 4th quantile was largest for CCL2/MCP1 (HR 1.69), followed by sTNFR2 (HR 1.61), CCL15/MIP1D (HR 1.39) and CCL27/CTACK (HR 1.35). Elevated levels of CCL3/MIP1A and CXCL6/GCP2 were associated with decreased risk with HRs for the 4th quantile of 0.61 and 0.70, respectively. There was no correlation among these six markers, with Spearman coefficient Rho ranging from -0.15 to 0.20 . The joint model of the selected six markers adjusted for other covariates did not alter the magnitude of association in each category, most of which stayed less than or close to 10% estimate change, with exceptions that associations became stronger for CCL15/MIP1D 2nd quartile (HR 1.13 to 1.39; 23% change) and CXCL6/GCP2 4th quartile (HR 0.70 to 0.48; 31% change). A complete list of CRC associations for all 62 detectable markers is provided in Supporting Information TableS2.

Table 3 shows a lag analysis of the 6 significantly associated markers. When restricting the analysis to cases diagnosed 5+ years after blood draw, CCL2/MCP1, CCL3/MIP1A and CXCL6/GCP2 retained nominal statistical significance that did not survive FDR correction.

Comparing cases occurring < 5 years vs. ≥ 5 years after the blood was collected, there was no significant difference for any of the five chemokines (all $P_{interaction} > 0.05$). On the other hand, $P_{interaction}$ was significant for sTNFR2 with a statistically significant association only for cases within the first 5 years.

An analysis comparing colon and rectum cancer subsites is presented in Table 4. Four of the six markers were nominally associated with either colon or rectum cancer, but none of the six subsite differences were statistically significant, with all $P_{heterogeneity} > 0.05$.

DISCUSSION

In this first comprehensive assessment of 62 circulating inflammatory markers and prospective CRC risk, five chemokines and a soluble cytokine receptor were each independently associated with the disease risk. The chemokines retained associations even after restricting to cases occurring ≥ 5 years after blood draw, indicating reverse causation is unlikely. Associations with colon and rectal cancers were similar.

Among the six markers found to be significantly associated with CRC risk, five were chemokines. Chemokines were originally identified as a class of cytokines that act as potent attractants for white blood cells to sites of infection or injury, acting as mediators of acute and chronic inflammation²⁰. To date, over 50 human chemokines have been discovered, categorized into four families (CC, CXC, CX3C and (X)C chemokines) based on their structure. They interact with a corresponding set of over 20 G-protein-coupled receptors in redundant and pleiotropic relationships^{20, 21}. With regards to carcinogenesis, chemokines may recruit tumor-associated leukocytes which induce angiogenesis and control tumor cell invasion, or directly induce migration of epithelial/endothelial cells that express their receptors²². Studies of chemokines in colorectal carcinogenesis are primarily limited to analyses of human tumors or mouse models. The few previous epidemiologic studies have generally assessed post-diagnosis blood samples from CRC cases and non-cancer controls, a design that determines markers of prevalent disease rather than associations with risk.

Given their pleiotropic activity, increased levels of chemokines could either promote or inhibit carcinogenesis^{23, 24}. We found elevated levels of 3 CC chemokines (CCL2/MCP1, CCL15/MIP1D and CCL27/CTACK) and decreased level of 1 CC chemokine (CCL3/MIP1A) and 1 CXCL chemokine (CXCL6/GCP2) to be associated with increased CRC risk.

Four of our five findings for circulating chemokine levels were consistent with prior studies of CRC tumor tissue and/or animal models. CCL2/MCP1 has been previously reported to be increased in colorectal tumor tissue²⁵, with higher expression associated with advanced tumor stage²⁶; knockout in a mouse model blocked progression from dysplasia to adenoma²⁷. With regards to CCL15/MIP1D, expression in liver metastases was associated with shorter disease-free survival²⁸; knockout of its receptor C-C chemokine receptor 1 (CCR1) blocked metastasis in a mouse model²⁹. CCL27/CTACK has also been found to have higher expression in tumors than paratumorous colorectal tissues³⁰. While CXCL6/GCPS expression was not altered in CRC tumors overall³¹, expression was lower in primary tumors of patients with disseminated CRC than in tumors that had not metastasized³².

Although we found CCL3/MIP1A to be inversely associated with CRC risk, knockout of this gene induced resistance to chemically induced colonic inflammation and tumors in a mouse model ³³.

Our finding for sTNFR2 is consistent with most prior epidemiologic studies of circulating levels. TNF is a pro-inflammatory cytokine with two main cognate receptors; expression of TNFR2 is restricted to certain cell types whereas TNFR1 is more ubiquitous. A case-control study reported higher levels of serum sTNFR2 in CRC cases than patients with benign disorders ³⁴. Prospective studies have had mixed results, with one study reporting a positive association with high levels and protection by aspirin/NSAID use that was restricted to individuals with high levels ³⁵, while two other studies were null ^{36, 37}.

BMI is a recognized risk factor for CRC ³⁸. There is growing evidence that obesity induces important physiologic changes collectively referred to as the inflammatory syndrome, which includes alterations in circulating levels of certain cytokines, chemokines and other inflammation mediators ^{39, 40}. We therefore adjusted for BMI so that our marker associations would not merely represent surrogates of obesity. Indeed, none of the six markers identified in the current analysis overlap with markers for BMI in this population (manuscript in preparation).

Our lag analysis showing consistency of the chemokine associations over the entire study period strengthened our interpretation of an etiologic role in colorectal carcinogenesis. On the other hand, the relative restriction of our sTNFR2 association to < 5 years after blood draw is more representative of an early disease marker, although with limited discriminatory power. The similarity of the associations for the colon and rectum implies common pathophysiology for the cancers of these sites.

The major strengths of our study include the population-based design, large number of well-characterized cancer cases, and detailed adjustment for potential confounders. The blood samples had been collected over the full spectrum of prediagnostic intervals for biologically relevant effects. We used highly reproducible state-of-the-art detection methods to assess a comprehensive set of inflammatory markers, with all but four markers having quantifiable levels in at least half of the samples. The case and subcohort samples were assayed contemporaneously, insuring consistency of the measurements.

Our study had some limitations that primarily would cause bias towards the null. The associations we found were relatively modest and not significant after FDR correction; power was limited for analyses by lag period and anatomic subsite. The studied markers tend to be labile, and measurements may be affected by degradation during processing, storage and testing ^{41, 42}. However, our samples had only been thawed and refrozen twice before the current analysis, and bead-based assays are generally unaffected by up to three freeze-thaw cycles ⁴³. We had only a single plasma sample from each subject, and it is uncertain how much an individual's levels vary over time. Circulating concentrations may not precisely reflect tissue-specific levels at the relevant site of action ⁴⁴, and blood measurements may only provide a summary measure of inflammatory activity within an individual ⁴⁵. Nevertheless, our results closely mirror previous findings for target tissues, suggesting

correlation of local and systemic inflammatory status. Moreover, we were not able to adjust for use of nonsteroidal anti-inflammatory drugs or other medications because we did not collect the relevant information.

In conclusion, our prospective study identified several potential blood markers for CRC risk, extending observations from laboratory studies and single-marker associations. Despite strong biological plausibility and consistency with previous studies, our study still requires replication of the specific associations. Further studies are warranted to confirm that chemokines may predict CRC risk.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

ACKNOWLEDGEMENTS

This study was supported by the Intramural Research Program of the National Cancer Institute. We gratefully acknowledge statistical assistance from Michael Curry, Information Management Services, Inc.

We are indebted to the Ibaraki, Niigata, Osaka, Kochi, Nagasaki, and Okinawa Cancer Registries for providing their incidence data. Members of the Japan Public Health Center-based Prospective Study (JPHC Study; Principal Investigator, S. Tsugane) Group (as of April 2017) are listed at <http://epi.ncc.go.jp/en/jphc/781/7951.html>.

Abbreviations Used:

BMI	body mass index
CCL	C-C motif ligand
CCR	C-C chemokine receptors
CI	confidence interval
CRC	colorectal cancer
CTACK	cutaneous T-cell attracting chemokine
CV	coefficient variation
CXCL	C-X-C motif ligand
GCP	granulocyte chemotactic protein
HR	hazard ratio
ICC	intra-class correlation
ICO-O-3	International Classification of Diseases for Oncology, 3 rd edition
IL	interleukin
IRB	Institutional Review Board

JPHC Study	Japan Public Health Center-based Prospective Study
LLOD	limit of detection
MCP	monocyte chemoattractant protein
MIP	macrophage inflammatory protein
NSAIDs	non-steroidal anti-inflammatory drugs
TNFR	tumor necrosis factor receptor

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Novelty and impact:

Previously, associations with colorectal cancer risk are based on few prospective studies with limited sets of markers. Our comprehensive analysis of 62 circulating markers found five were associated with colorectal cancer. Chemokines CCL2/MCP1, CCL3/MIP1A and CXCL6/GCP2 retained nominal statistical significance restricting to cases ≥ 5 years after enrollment. There were no significant differences in marker associations between colon and rectum cancers.

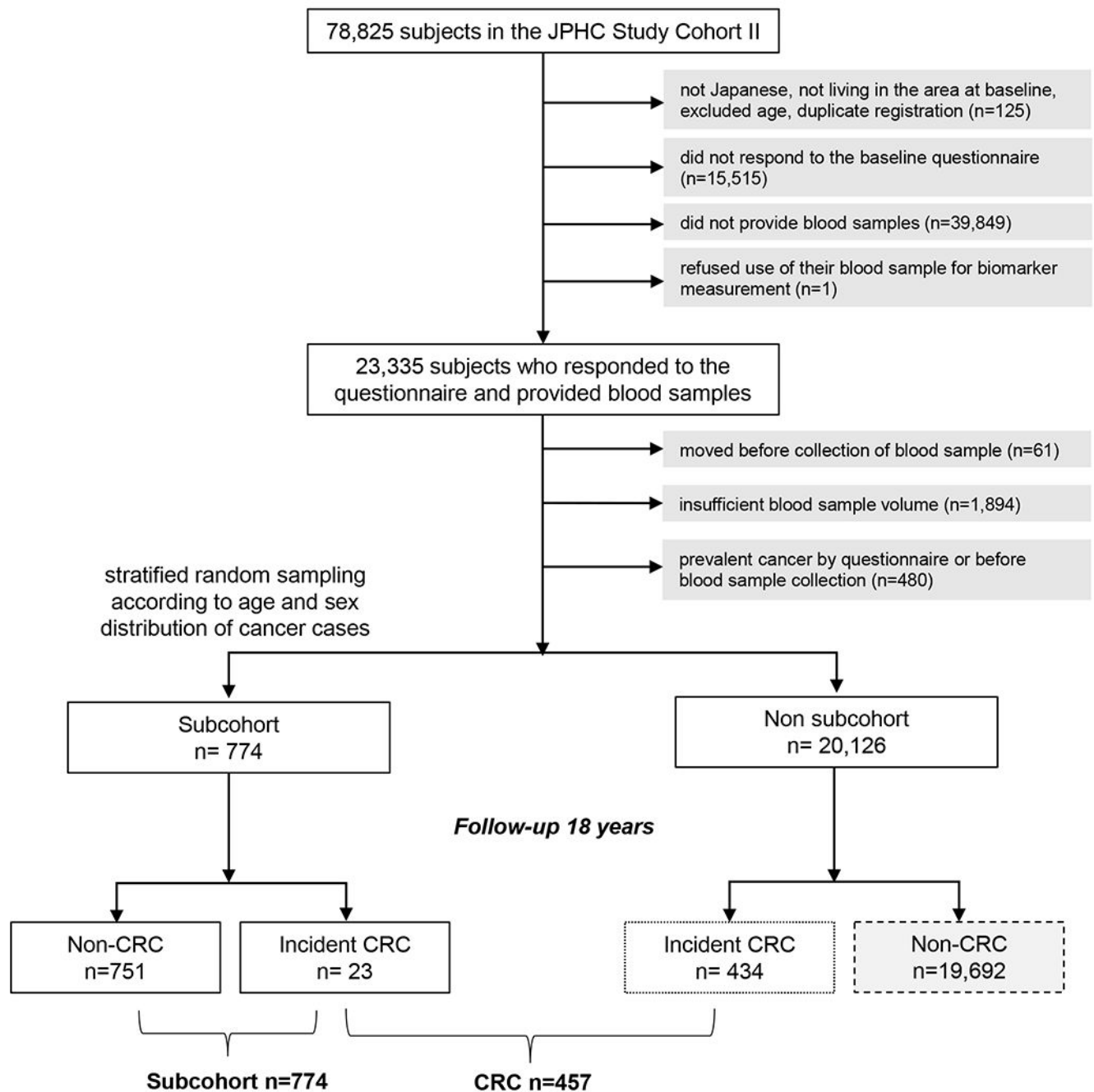


Figure 1.

Case-cohort sample design nested within the JPHC Study.

Flow diagram of selection of subcohort and incident CRC cases from a total of 78,825 participants in the JPHC Study Cohort II. After accounting for exclusion criteria, stratified random sampling was performed to obtain a comparison group of 774 individuals with similar age and sex distribution as cancer cases. There were a total of 457 CRC cases of which 23 occurred among members of the subcohort.

Abbreviations: CRC = colorectal cancer, JPHC Study = Japan Public Health Center-based Prospective Study

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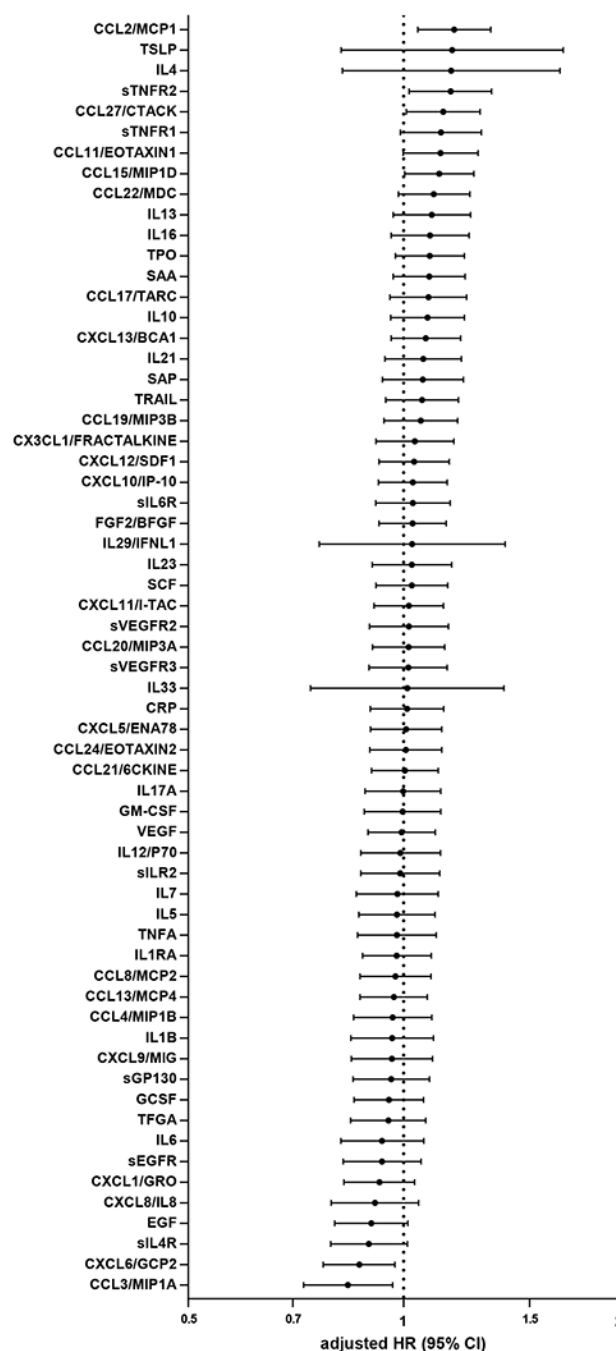


Figure 2.

Association of inflammatory markers with incident colorectal cancer in the JPHC study. Cox proportional hazards model estimated hazard ratios (HR) and 95% confidence intervals (95% CI) for colorectal cancer risk per quantile increase of inflammatory markers. Models were adjusted for age (in years), gender and study area (6 public health center areas), family history of colorectal cancer (yes or no), history of diabetes (yes or no), BMI category (< 21, 21-22.9, 23-26.9, ≥ 27 kg/m²), smoking habits (never, past, current [< 20 and ≥ 20 cigarettes/day]), alcohol drinking (no or occasionally drinking, currently drinking almost

everyday [1-150, 151-300 and $\geq$ 300g ethanol/week]) and physical activity (quartiles of metabolic equivalents/day).

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Table 1.

Baseline characteristics of colorectal cancer case patients and subcohort members of JPHC

	Subcohort (n=774)		Colorectal cancer (n=457)	
Age At Enrollment, mean $\pm$ SD, years	60.1 $\pm$ 7.5		60.3 $\pm$ 7.2	
Male, %	413	(53.4%)	219	(47.9%)
Family Hx of Colorectal Cancer, %*	15	(1.9%)	11	(2.4%)
Smoking				
Never	452	(58.4%)	275	(60.2%)
Former	134	(17.3%)	74	(16.2%)
Current $\leq$ 20 cig/day	129	(16.7%)	74	(16.2%)
Current >20 cig/day	47	(6.1%)	29	(6.3%)
Unknown	12	(1.6%)	5	(1.1%)
Alcohol Drinking, %				
Never/occasional	552	(71.3%)	313	(68.5%)
1+ per day and <300 g/week	135	(17.4%)	77	(16.8%)
1+ per day and =300 g/week	64	(8.3%)	50	(10.9%)
Unknown	23	(3.0%)	17	(3.7%)
BMI, %				
<21kg/m ²	155	(20.0%)	81	(17.7%)
21-22.9 kg/m ²	189	(24.4%)	112	(24.5%)
23-26.9 kg/m ²	331	(42.8%)	203	(44.4%)
$\geq$ 27 kg/m ²	87	(11.2%)	53	(11.6%)
Unknown	12	(1.6%)	8	(1.8%)
Physical Activity (METs/day), %				
$\geq$ 42.7	160	(20.7%)	121	(26.5%)
34.3-42.6	148	(19.1%)	63	(13.8%)
31.9-34.2	178	(23.0%)	94	(20.6%)
< 31.8	259	(33.5%)	160	(35.0%)
Unknown	29	(3.7%)	19	(4.2%)
History of Diabetes, %				
No	311	(40.2%)	190	(41.6%)
Yes	46	(5.9%)	36	(7.9%)
Unknown	417	(53.9%)	231	(50.5%)

Table 2.

Circulating inflammation markers statistically significantly associated with colorectal cancer risk *

		Individual Model		Joint Model
Marker and level, pg/mL	Case (n)	HR1 [†] (95% CI)	HR2 [‡] (95% CI)	HR3 [§] (95% CI)
CCL2/MCP1				
≤445.38	98	1.00	1.00	1.00
445.40-549.69	110	1.22 (0.86-1.75)	1.21 (0.83-1.75)	1.10 (0.70-1.75)
549.89-674.99	116	1.34 (0.94-1.90)	1.26 (0.87-1.82)	1.47 (0.90-2.39)
≥675.25	132	1.71 (1.20-2.45)	1.69 (1.17-2.45)	1.55 (0.92-2.62)
P _{trend}		0.003	0.007	0.07
CCL3/MIP1A				
≤28.96	113	1.00	1.00	1.00
28.99-43.08	100	0.90 (0.63-1.28)	0.94 (0.64-1.39)	1.11 (0.70-1.78)
43.14-75.13	93	0.67 (0.46-0.98)	0.68 (0.46-1.02)	0.69 (0.43-1.12)
≥75.15	84	0.58 (0.38-0.89)	0.61 (0.39-0.95)	0.68 (0.39-1.18)
P _{trend}		0.007	0.02	0.05
CCL15/MIP1D				
≤2446.43	102	1.00	1.00	1.00
2450.04-3384.02	106	1.09 (0.77-1.55)	1.13 (0.78-1.64)	1.39 (0.86-2.26)
3384.87-5308.52	124	1.26 (0.90-1.76)	1.33 (0.94-1.90)	1.47 (0.93-2.31)
≥5313.51	124	1.39 (0.99-1.95)	1.39 (0.97-1.98)	1.32 (0.82-2.13)
P _{trend}		0.04	0.04	0.14
CCL27/CTACK				
≤589.96	101	1.00	1.00	1.00
590.3-726.74	106	1.00 (0.71-1.42)	1.03 (0.72-1.48)	0.99 (0.62-1.59)
727.28-888.33	133	1.42 (1.01-2.01)	1.48 (1.03-2.14)	1.74 (1.09-2.78)
≥888.69	116	1.33 (0.93-1.91)	1.35 (0.92-1.97)	1.31 (0.80-2.15)
P _{trend}		0.03	0.04	0.06
CXCL6/GCP2				
≤45.70	124	1.00	1.00	1.00
45.74-61.60	137	1.07 (0.78-1.48)	1.14 (0.82-1.60)	1.11 (0.72-1.70)
61.69-80.66	95	0.71 (0.50-0.99)	0.75 (0.53-1.08)	0.73 (0.46-1.15)
≥80.75	97	0.65 (0.46-0.93)	0.70 (0.48-1.02)	0.48 (0.29-1.82)
P _{trend}		0.003	0.02	0.004
sTNFR2				
≤6418.29	85	1.00	1.00	1.00
6428.68-7754.05	92	1.05 (0.92-1.55)	1.04 (0.69-1.55)	1.13 (0.71-1.80)
7756.06-9379.25	90	1.00 (0.69-1.47)	1.00 (0.67-1.49)	1.06 (0.66-1.71)
≥9380.91	125	1.47 (1.00-2.16)	1.61 (1.07-2.41)	1.80 (1.12-2.91)
P _{trend}		0.06	0.03	0.04

* Cox proportional hazards model stratified by age group and gender was used to estimate hazard ratios (HRs) and 95% confidence intervals (CIs). Abbreviations: CCL=chemokine (C-C motif) ligand, CXCL=chemokine (C-X-C motif) ligand, CTACK=cutaneous T-cell attracting chemokine, GCP2=granulocyte chemotactic protein 2, MCP1=monocyte chemoattractant protein 1, MIP1A=macrophage inflammatory protein 1A, MIP1D=macrophage inflammatory protein 1D, sTNFR2=soluble necrosis factor receptor 2

[†]HR1 was adjusted for age (in years), gender and study area (6 public health center areas)

[‡]HR2 was adjusted for covariates used in HR1 and family history of colorectal cancer (yes or no), history of diabetes (yes or no), BMI category (<21, 21-22.9, 23-26.9, and ≥ 27 kg/m²), smoking habits (never, past, current [<20 and ≥ 20 cigarettes/day]), alcohol drinking (no or occasionally drinking, currently drinking almost everyday [1-150, 151-300 and ≥ 300 g ethanol/week]), physical activity (quartiles of metabolic equivalents/day)

[§]HR3 was adjusted for the same covariates as HR2, and the other significant markers as categorical variables for categorical HR (95% CI)

Table 3.

Circulating inflammation markers statistically significantly associated with colorectal cancer risk in different phases *

Marker and level, pg/mL	Years from sample until diagnosis				$P_{interaction}^{\ddagger}$
	< 5 years (n=125)		≥ 5 years (n=332)		
	Case (n)	HR [†] (95% CI)	Case (n)	HR [†] (95% CI)	
CCL2/MCP1					
≤445.38	22	1.00	76	1.00	0.31
445.40-549.69	33	1.45 (0.88-2.41)	77	1.13 (0.76-1.68)	
549.89-674.99	29	1.33 (0.79-2.23)	87	1.24 (0.84-1.83)	
≥675.25	41	2.04 (1.25-3.33)	91	1.57 (1.06-2.33)	
P_{trend}		0.005		0.02	
CCL3/MIP1A					
≤28.96	32	1.00	81	1.00	0.64
28.99-43.08	25	0.83 (0.48-1.44)	75	0.99 (0.65-1.49)	
43.14-75.13	18	0.49 (0.27-0.89)	75	0.76 (0.50-1.15)	
≥75.15	24	0.62 (0.35-1.10)	60	0.60 (0.37-0.97)	
P_{trend}		0.02		0.02	
CCL15/MIP1D					
≤2446.43	25	1.00	77	1.00	0.61
2450.04-3384.02	28	1.24 (0.73-2.10)	78	1.10 (0.74-1.63)	
3384.87-5308.52	38	1.65 (1.01-2.71)	86	1.23 (0.85-1.79)	
≥5313.51	34	1.39 (0.85-2.27)	90	1.40 (0.95-2.05)	
P_{trend}		0.05		0.07	
CCL27/CTACK					
≤586.96	24	1.00	77	1.00	0.70
590.34-726.74	34	1.32 (0.81-2.15)	72	0.93 (0.63-1.38)	
727.28-888.33	37	1.60 (0.96-2.64)	96	1.45 (0.99-2.13)	
≥888.69	30	1.40 (0.83-2.36)	86	1.33 (0.89-2.00)	
P_{trend}		0.05		0.06	
CXCL6/GCP2					
≤45.70	30	1.00	94	1.00	0.18
45.74-61.60	38	1.25 (0.79-1.98)	99	1.11 (0.78-1.58)	
61.69-80.66	24	0.89 (0.47-1.33)	71	0.74 (0.51-1.09)	
≥80.75	32	0.93 (0.57-1.53)	65	0.63 (0.42-0.94)	
P_{trend}		0.24		0.007	
sTNFR2					
≤6418.29	14	1.00	71	1.00	0.01
6428.68-7754.05	24	1.36 (0.77-2.39)	68	0.96 (0.63-1.47)	
7756.06-9379.25	28	1.46 (0.85-2.50)	62	0.88 (0.57-1.36)	
≥9380.91	43	2.42 (1.44-4.05)	82	1.40 (0.91-2.15)	

Years from sample until diagnosis					
		< 5 years (n=125)		≥ 5 years (n=332)	
Marker and level, pg/mL	Case (n)	HR [†] (95% CI)	Case (n)	HR [†] (95% CI)	P _{interaction} [‡]
P _{trend}		0.001		0.12	

* Cox proportional hazards model stratified by age group and gender was used to estimate hazard ratios (HRs) and 95% confidence intervals (CIs). Abbreviations: CCL=chemokine (C-C motif) ligand, CXCL=chemokine (C-X-C motif) ligand, CTACK=cutaneous T-cell attracting chemokine, GCP2=granulocyte chemotactic protein 2, MCP1=monocyte chemoattractant protein 1, MIP1A=macrophage inflammatory protein 1A, MIP1D=macrophage inflammatory protein 1D, sTNFR2=soluble necrosis factor receptor 2

[†]HR was adjusted for age (in years), gender, study area (6 public health center areas), family history of colorectal cancer (yes or no), history of diabetes (yes or no), BMI category (<21, 21-22.9, 23-26.9, and ≥27 kg/m²), smoking habits (never, past, current [<20 and ≥20 cigarettes/day]), alcohol drinking (no or occasionally drinking, currently drinking almost everyday [1-150, 151-300 and ≥300g ethanol/week]), physical activity (quartiles of metabolic equivalents/day)

[‡]P_{interaction} calculated based on ordinal quantile levels of markers, adjusted for the same covariates as HR1

Table 4.

Circulating inflammation markers statistically significantly associated with colorectal cancer risk in different sites*

	Colon (n=319)		Rectum (n=149)		
Marker and level, pg/mL	Case (n)	HR [†] (95% CI)	Case (n)	HR [†] (95% CI)	P _{heterogeneity} [‡]
CCL2/MCP1					
≤445.38	68	1.00	33	1.00	0.15
445.40-549.69	70	1.12 (0.73-1.71)	42	1.27 (0.72-2.23)	
549.89-674.99	78	1.21 (0.79-1.84)	39	1.27 (0.74-2.20)	
≥675.25	102	1.95 (1.29-2.94)	34	1.20 (0.67-2.15)	
P _{trend}		0.002		0.53	
CCL3/MIP1A					
≤28.96	76	1.00	43	1.00	0.24
28.99-43.08	71	1.06 (0.68-1.65)	31	0.71 (0.41-1.23)	
43.14-75.13	63	0.69 (0.44-1.09)	30	0.57 (0.31-1.03)	
≥75.15	63	0.68 (0.41-1.13)	22	0.43 (0.22-0.84)	
P _{trend}		0.06		0.01	
CCL15/MIP1D					
≤2446.43	72	1.00	32	1.00	0.68
2450.04-3384.02	74	1.18 (0.77-1.82)	36	1.11 (0.64-1.94)	
3384.87-5308.52	90	1.45 (0.78-2.15)	36	1.10 (0.63-1.90)	
≥5313.51	82	1.33 (0.89-1.99)	45	1.47 (0.86-2.52)	
P _{trend}		0.10		0.18	
CCL27/CTACK					
≤586.96	73	1.00	30	1.00	0.76
590.34-726.74	73	0.99 (0.66-1.50)	36	1.11 (0.64-1.94)	
727.28-888.33	92	1.45 (0.96-2.18)	44	1.55 (0.89-2.70)	
≥888.69	80	1.27 (0.83-1.95)	39	1.42 (0.78-2.57)	
P _{trend}		0.11		0.14	
CXCL6/GCP2					
≤45.70	87	1.00	41	1.00	0.23
45.74-61.60	87	1.04 (0.71-1.52)	52	1.35 (0.83-2.18)	
61.69-80.66	68	0.77 (0.51-1.15)	29	0.70 (0.40-1.22)	
≥80.75	74	0.74 (0.49-1.12)	25	0.59 (0.32-1.07)	
P _{trend}		0.07		0.02	
sTNFR2					
≤6418.29	59	1.00	29	1.00	0.25
6428.68-7754.05	58	0.92 (0.57-1.49)	37	1.27 (0.73-2.34)	
7756.06-9379.25	68	1.07 (0.68-1.68)	23	0.73 (0.38-1.42)	
≥9380.91	91	1.76 (1.12-2.77)	37	1.25 (0.67-2.32)	
P _{trend}		0.009		0.87	

* Cox proportional hazards model stratified by age group and gender was used to estimate hazard ratios (HRs) and 95% confidence intervals (CIs). Abbreviations: CCL=chemokine (C-C motif) ligand, CXCL=chemokine (C-X-C motif) ligand, CTACK=cutaneous T-cell attracting chemokine, GCP2=granulocyte chemotactic protein 2, MCP1=monocyte chemoattractant protein 1, MIP1A=macrophage inflammatory protein 1A, MIP1D=macrophage inflammatory protein 1D, sTNFR2=soluble tumor necrosis factor receptor 2

[†]HR was adjusted for age (in years), gender, study area (6 public health center areas), family history of colorectal cancer (yes or no), history of diabetes (yes or no), BMI category (<21, 21-22.9, 23-26.9, and ≥27 kg/m²), smoking habits (never, past, current [<20 and ≥20 cigarettes/day]), alcohol drinking (no or occasionally drinking, currently drinking almost everyday [1-150, 151-300 and ≥300g ethanol/week]), physical activity (quartiles of metabolic equivalents/day)

[‡]*Pheterogeneity* calculated based on ordinal quantile levels of markers, adjusted for the same covariates as HR1